ANOVA 1

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Topics for this lecture

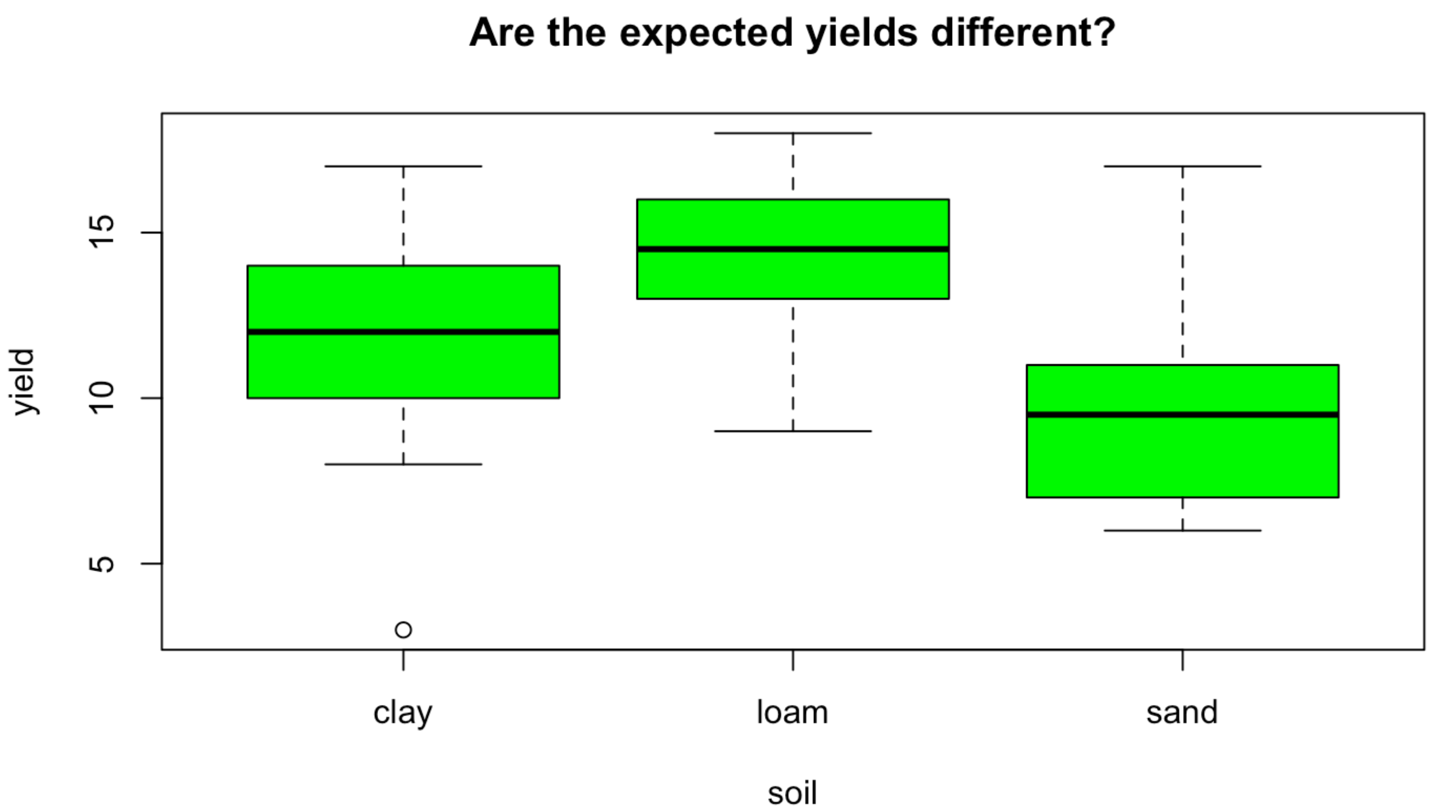
- Introductory example
- “ANOVA”
- The F-test
- Estimation and parametrization
- Post hoc contrasts and pairwise tests
- Errors and power

Objectives of lecture

- *Basic framework*: The response variable is continuous as for regression but the predictor variables are *factors* like fertilizer, soil type, country, …
- *Basic idea*: The statistical analysis is based on studying the variability in data. Hence the name ANOVA:
  - ANalysis Of VAriance.
- Objectives as for regression: predict and explain

Example Yields

Yield from a crop experiment on $k = 3$ different soil types. There are $n = 10$ random replicates per soil type. A total of $N = k \cdot n = 30$ observations altogether.



The model

As in regression analysis we assume a linear model with both an explainable part and an unexplainable part (noise)

Let y_{ij} be the observed yield for observation number $j \in \{1, \dots, n_i\}$ soil type $i \in \{1, \dots, k\}$.

The model (One-way ANOVA):

$$y_{ij} = \mu_i + \epsilon_{ij}$$

where $\epsilon_{ij} \sim N(0, \sigma^2)$ and μ_i is the expected yield for soil type i.

Splitting up total variability

SST= $\sum_{i=1}^{k} \sum_{j=1}^{n} (y_{ij} - \bar{y}_{..})^2, \quad \bar{y}_{..}$ is grand mean

For the yield data:

```
## [1] 414.7
```

SSA =
$\sum_{i=1}^{k} \sum_{j=1}^{n} (\bar{y}_{i.} - \bar{y}_{..})^2$ = $n \sum_{i=1}^{k} (\bar{y}_{i.} - \bar{y}_{..})^2, \quad \bar{y}_{i.}$ i-th group mean

For the yield data:

```
## [1] 99.2
```

- SSE = SST - SSA = $\sum_{i=1}^{k} \sum_{j=1}^{n} (y_{ij} - \bar{y}_{i.})^2$ = 414.7 - 99.2 = 315.5

Between and within group variances:

Dividing Sums of Squares by their respective degrees of freedom gives:

- The between soil type (group) variance:
  $MSA = SSA/(k-1) = 99.2/2 = 49.6$
- The within soil type (group) variance:
  $MSE = SSE/(n-k) = 315.5/27 = 11.685$

where $n = n_1 + n_2 + \cdots n_k$ is the total number of observations.

Why does it make sense to compare these?

Testing the effect of soil

If there is a soil effect on yield, the expected yields will differ in one way or the other.

Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

vs

$$H_1 : \mu_i \neq \mu_{i'}$$

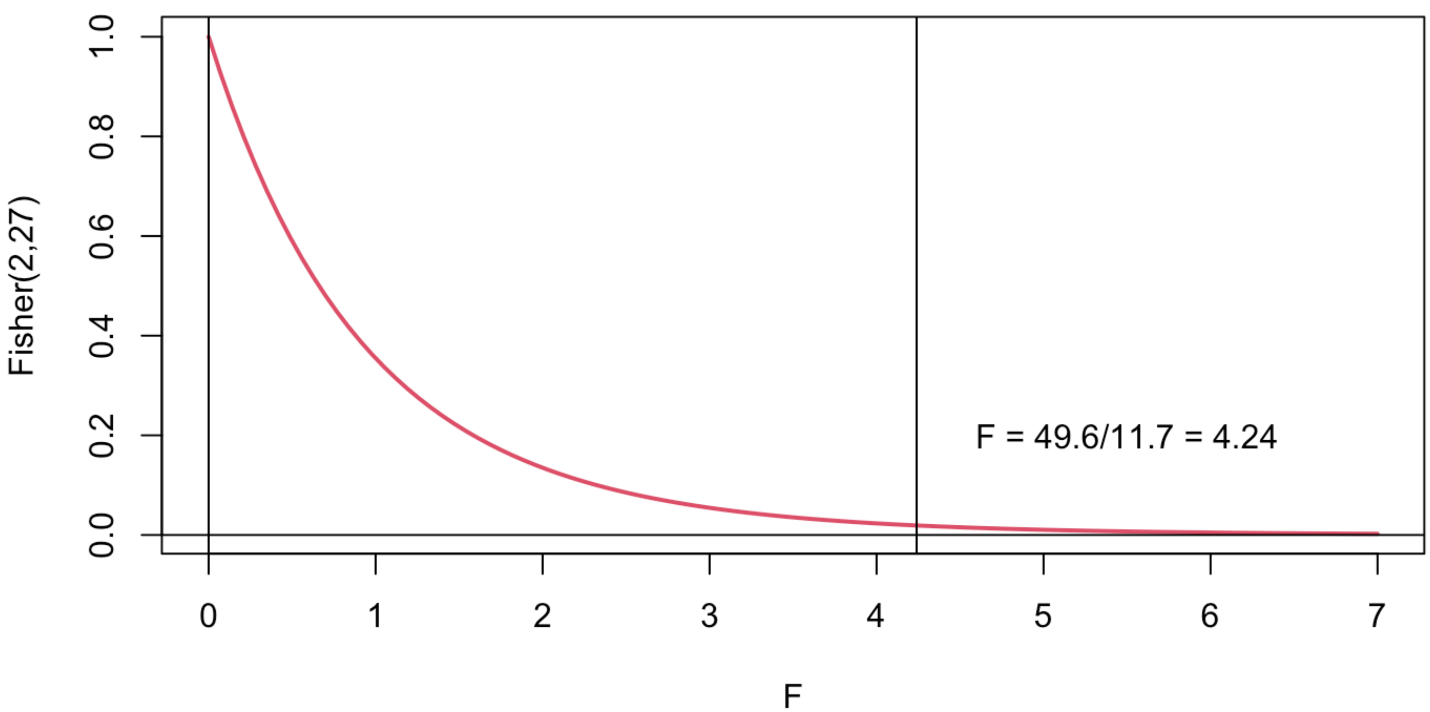
for at least two different soil types i and i'.

Reject H_0 if $MSA >> MSE$

Test observator, F

If H_0 is true, then

$$F = \frac{MSA}{MSE} \sim F(\nu_1 = k - 1 = 3 - 1, \nu_2 = n - k = 30 - 3)$$



Rejection region

Reject H_0 at test level α if $F_{obs} > F_{\alpha,\nu_1,\nu_2}$, where $\nu_1 = k - 1$ are the degrees of freedom for MSA and $\nu_2 = n - k$ are the degrees of freedom for MSE.

For example, assume a test level of 5%

From a Fisher table we find $F_{0.05,2,27}$ = 3.35

Alternatively, reject if p-value is smaller than α.

Conclusion?

Estimation of group effects and error variance

The inutitive estimators for μ_i are $\widehat{\mu}_i = \bar{y}_{i.}$

These are also the least squares estimators (for balanced data), which are unbiased and have minimum variance among all unbiased estimators.

As in regression analysis we have $\hat{\sigma}^2 = MSE$.

Splitting up μ_i - i.e. reparametrise

The means may be split up further. Assume

$$\mu_i = \mu + \alpha_i$$

But, all parameters can not be estimated without some extra restriction (choice of parametrization).

- Parametrization 1: Assume $\sum_{i=1}^{k} \alpha_i = 0$

A "sum to zero" parametrization ("contr.sum" in R")

- Parametrization 2: Assume $\alpha_1 = 0$

A "reference level" or "control level" parametrization ("contr.treatment" in R")

Choice of arametrization in RStudio

Parametrization 1 (Using the lm-function)

```
mod1 <- lm(yield~soil, data=yields, contrasts = "contr.sum")
```

Parametrization 2 (Using the lm-function)

```
mod1 <- lm(yield~soil, data=yields, contrasts = "contr.treatment")
```

Example: Parametrization 1

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```
##             Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)     11.5   1.080980 10.638491 3.702061e-11
## soilloam         2.8   1.528737  1.831577 7.806971e-02
## soilsand        -1.6   1.528737 -1.046616 3.045565e-01
```

Compare with group and total means:

```
means <- c(soilmeans,mean(soilmeans))
names(means) <- c("clay", "loam", "sand", "total-mean")
print(means)
```

```
##       clay       loam       sand total-mean
##       11.5       14.3        9.9       11.9
```

Example: Parametrization 2

```
## Warning in model.matrix.default(mt, mf, contrasts): non-list contrasts argument
## ignored
```

```
##             Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)     11.5   1.080980 10.638491 3.702061e-11
## soilloam         2.8   1.528737  1.831577 7.806971e-02
## soilsand        -1.6   1.528737 -1.046616 3.045565e-01
```

Compare with group and total means:

```
##       clay       loam       sand total-mean
##       11.5       14.3        9.9       11.9
```

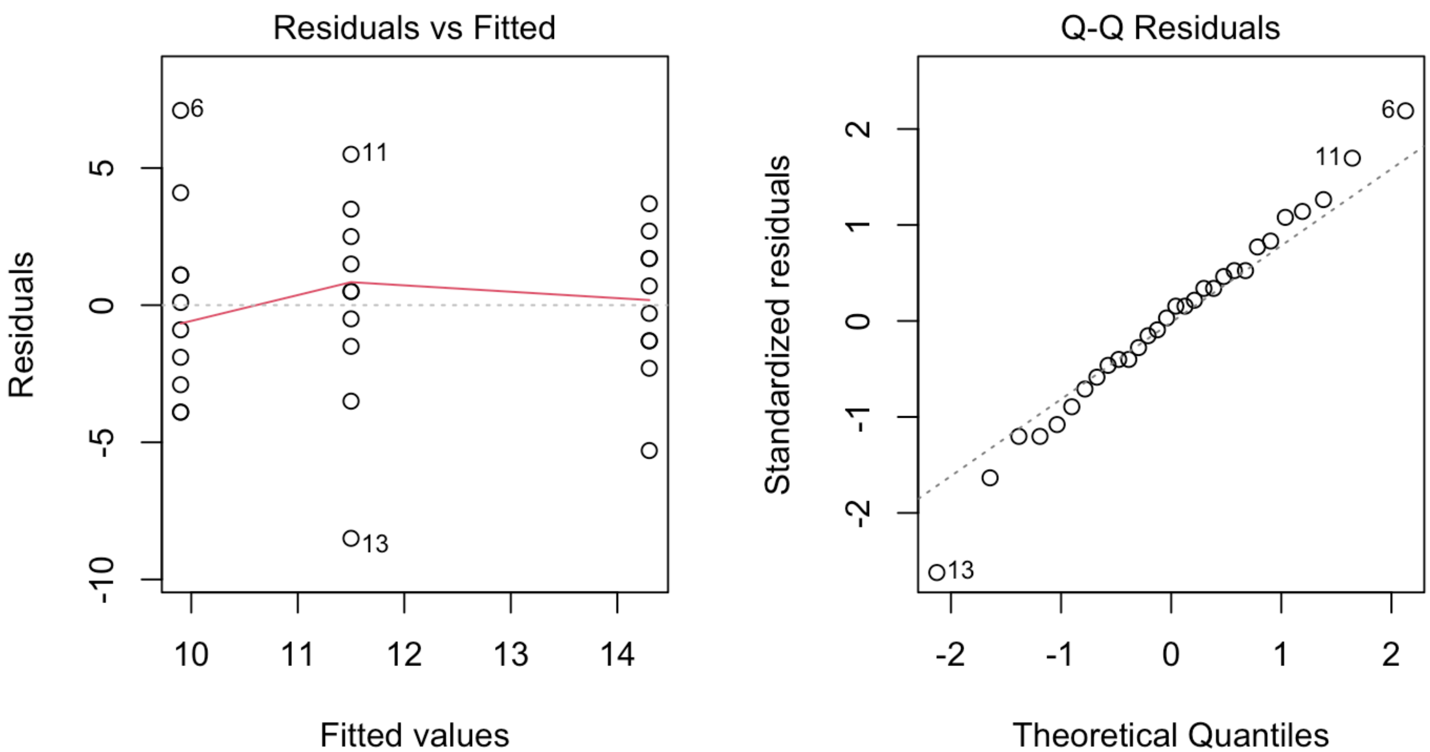
ANOVA table of Sum of squares

This table does not depend on parametrization

```
## Analysis of Variance Table
##
## Response: yield
##           Df Sum Sq Mean Sq F value  Pr(>F)
## soil       2   99.2  49.600  4.2447 0.02495 *
## Residuals 27  315.5  11.685
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual analysis

We should also for ANOVA-models check the assumptions by residual analysis



Pair-wise contrasts

If a factor is significant and has more than two levels: Where are the differences?

May perform a "post hoc" test with contrasts.

A contrasts is a linear function of the factor effects.

$$\theta = c_1\mu_1 + c_2\mu_2 + c_3\mu_3$$

where $\sum_{i=1}^{k} c_i = 0$

For pair-wise contrasts, only two c_i's differ from 0.

Example 1 (With parametrization 1)

Comparing soil type “loam” (level 1) with “clay” (level 2):

$$\theta_1 = 1\mu_1 + (-1)\mu_2 + 0\mu_3$$

$$= 1(\mu + \alpha_1) - 1(\mu + \alpha_2) = \alpha_1 - \alpha_2$$

Estimate: $\hat{\theta}_1 = \hat{\alpha}_1 - \hat{\alpha}_2 = -0.4 - 2.4 = -2.8$

```
options(contrasts=c("contr.sum","contr.poly"))
mod1 <- stats::lm(yield~soil, data=yields)
gmodels::fit.contrast(model=mod1, varname='soil', df=TRUE, coeff=c(1, -1, 0), conf.int
```

```
##                    Estimate Std. Error   t value   Pr(>|t|) DF  lower CI
## soil c=( 1 -1 0 )      -2.8   1.528737 -1.831577 0.07806971 27 -5.936709
##                     upper CI
## soil c=( 1 -1 0 ) 0.3367094
```

Example 2: “loam” vs “sand”

Define required pairwise contrast:

$$\theta_2 = 0\mu_1 + 1\mu_2 - 1\mu_3 = 1(\mu + \alpha_2) - 1(\mu + \alpha_3) = \alpha_2 - \alpha_3$$

Estimation

$$\widehat{\alpha}_3 = -(-0.4 + 2.4) = -2 \ \widehat{\theta}_2 = \widehat{\alpha}_2 - \widehat{\alpha}_3 = 2.4 - (-2) = 4.4$$

```
##                      Estimate Std. Error  t value    Pr(>|t|) DF lower CI upper CI
## soil c=( 0 1 -1 )       4.4   1.528737 2.878193 0.007728022 27 1.263291 7.536709
```

Multiple testing - Tukey test

Type 1: α = P(Rejecting H_0 | H_0 is TRUE)

If there are many levels of a factor, a Tukey test is preferable.

Controls overall error rate (family-wise error) of all pairwise tests.

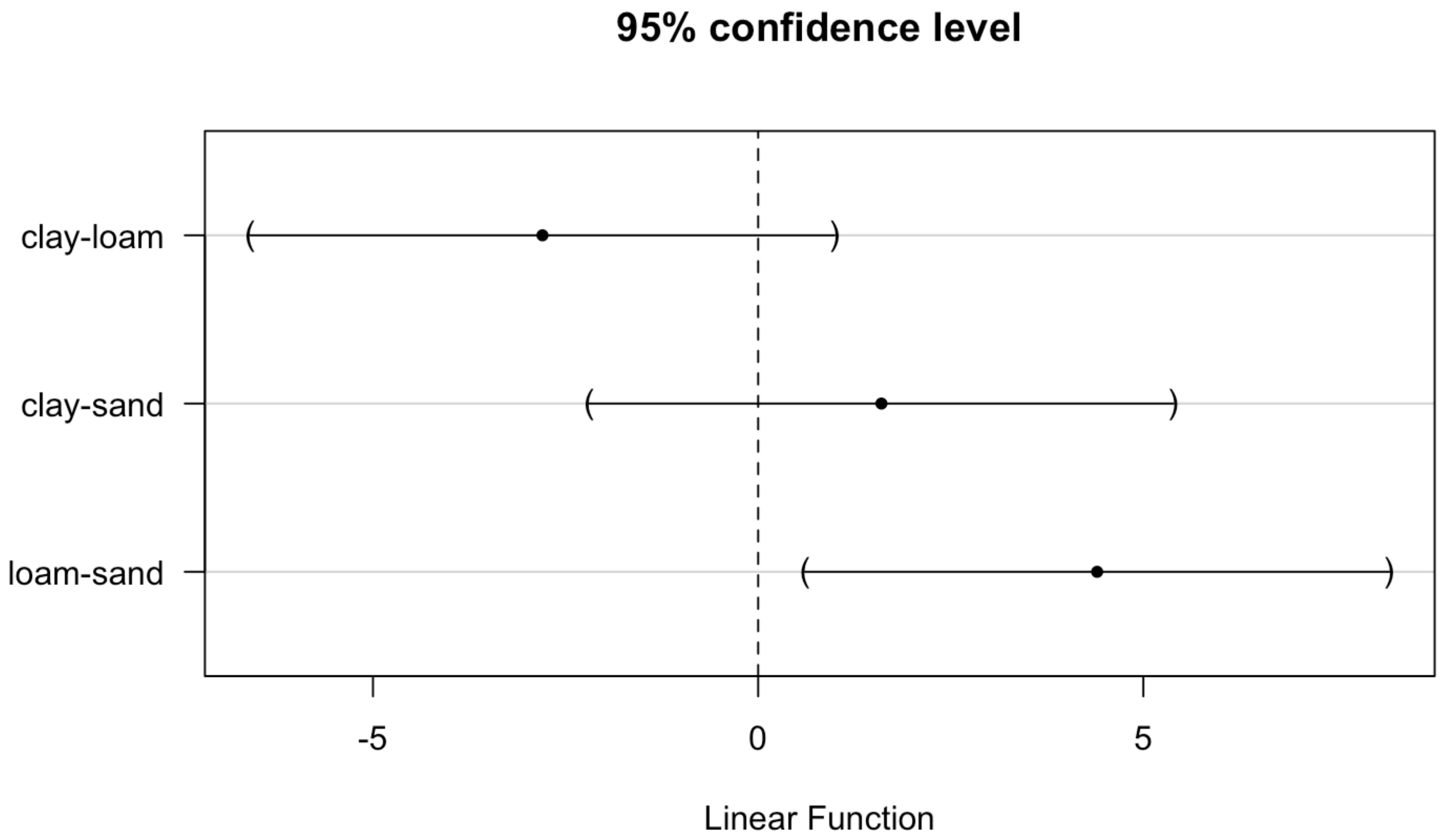
Use ‘simple.glht’-function in the mixlm-package.

```
library(mixlm)
ptests <- simple.glht(mod1, "soil", corr = c("Tukey"),level = 0.95)
print(ptests)
plot(ptests)
```

R Commander output - Tukey

```
##
##   Simultaneous Confidence Intervals and Tests for General Linear Hypotheses
##
## Multiple Comparisons of Means: Tukey Contrasts
##
##
## Fit: stats::lm(formula = yield ~ soil, data = yields)
##
## Quantile = 2.4794
## Minimum significant difference = 3.7904
## 95% confidence level
##
## Linear Hypotheses:
##             Lower  Center   Upper Std.Err t value  P(>t)
## clay-loam -6.5904 -2.8000  0.9904  1.5287  -1.832 0.1785
## clay-sand -2.1904  1.6000  5.3904  1.5287   1.047 0.5546
## loam-sand  0.6096  4.4000  8.1904  1.5287   2.878 0.0204 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## (Adjusted p values reported -- single-step method)
```

Pairwise confidence intervals plot



The ANOVA applet

The ANOVA “idea”: Studying means by splitting variability

ANOVA shiny app

Type 1 and 2 errors

We should keep in mind the errors which we may make

Type 1: α = P(Rejecting H_0 | H_0 is TRUE)

Type 2: β = P(Retaining H_0 | H_0 is FALSE)

Choosing the test level α always has an impact on β.

Test power

The test power is the ability to find existing effects:

Power: $(1 - \beta)$ = P(Rejecting H_0 | H_0 is FALSE)

Which factors influence the power for a given test?